Seat No. _____ Enrollment No. _____

THE CHARUTAR VIDYA MANDAL UNIVERSITY

B.E.– SEMESTER 6

APRIL 2024 EXAMINATION

Course Title: Smart Cities Planning and Management

Course Code: 102030622

Total Printed Pages: 2

Date: 03/05/2024 **Time: 10.00 am to 12.30 pm** **Maximum Marks: 60**

Instructions:

- Attempt all questions.
- Numbers to the right indicate full marks for each question.
- Make suitable assumption if required, do specify the same.

Q: 1 Answer the Following. (Each question carries one marks). **[10]**

(i) _____ are the part of SOCIAL infrastructure of a smart city.
[a] Hospitals [b] Water supply systems [c] Industrial parks [d] Financial centres

(ii) 'GIFT' stands for _____.
[a] Gujarat International Finance Technical City [b] Gujarat International Foreign Tec City [c] Gujarat International Finance Tec City [d] Gujarat Industrial Finance Tec City

(iii) 'Traffic management' is one of the features of Smart cities.
[a] true [b] false

(iv) "100 Smart cities mission" was launched in India in the year _____.
[a] 2014 [b] 2015 [c] 2016 [d] 2017

(v) Dahod, Ahmedabad, Surat, Vadodara, Rajkot and _____ were the cities selected from the state of Gujarat under the Smart city mission of India.
[a] Bhavnagar [b] Jamnagar [c] Anand [d] Gandhinagar

(vi) _____ will effect a replacement of the existing built-up environment and enable co-creation of a new layout with enhanced infrastructure using mixed land use and increased density.
[a] Retrofitting [b] Redevelopment [c] Pan city project [d] Green field development

(vii) Smart city projects not only contribute towards employment generation, but, also drive overall economic growth and enable the improvement in the quality of life of the citizens.
[a] true [b] false

(viii) Solid waste management systems, energy systems and urban mobility systems are part of _____ infrastructure of a smart city.
[a] Institutional [b] Physical [c] Social [d] Economic

(ix) SCADA stands for _____.
[a] Supervisory control and data acquiring [b] Supervisory control and data analysis [c] Supervisory control and data acquisition [d] Supervisory collection and data acquisition

(x) Smart city is defined as the effective integration of physical, digital and human systems in the built environment to deliver sustainable, prosperous and inclusive future of its citizens.
[a] true [b] false

Q: 2 **(A)** Explain the objectives / goals of smart cities. **[04]**

(B) Discuss any one case study of smart city. **[06]**

Q: 3 **(A)** Explain the challenges faced by the smart cities in India. **[05]**

(B) Write short note on: Smart city mission of India. **[05]**

OR

(B) Write short note on various infrastructure systems required for the smart city. **[05]**

Q: 4 **(A)** Enlist and explain the objectives of Green Buildings. **[05]**

(B) Write short note on: Green Rating for Integrated Habitat Assessment (GRIHA). **[05]**

OR

(B) Discuss the major benefits of solar energy. **[05]**

Q: 5 **(A)** Explain the criteria for housing for smart cities. **[05]**

(B) Explain in detail the objectives of Intelligent Transport System (ITS). **[05]**

OR

(B) Discuss the advantages of Rainwater Harvesting System (RWHS). **[05]**

Q: 6 **(A)** Describe various causes of floods in major rivers of India. **[05]**

(B) What is the scope of traffic safety management? Enumerate collective responsibilities of various segments of traffic safety management. **[05]**

OR

(B) Discuss various applications of Geographic Information System (GIS) in Transportation Engineering. **[05]**

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